

DOCK2 Deficiency — Comprehensive Disease Characterization

Disease: DOCK2 Deficiency **MONDO ID:** MONDO:0014637 **OMIM:** 616433 (Immunodeficiency 40, phenotype); 603122 (DOCK2 gene) **Orphanet:** ORPHA:317425 (Combined immunodeficiency due to DOCK2 deficiency) **Category:** Mendelian, autosomal recessive combined immunodeficiency **Report type:** Aggregated disease-level synthesis of primary literature (25 papers reviewed, 14 findings confirmed)

Summary

DOCK2 (Dedicator of Cytokinesis 2) deficiency is an ultra-rare, early-onset **autosomal recessive combined immunodeficiency (CID)** caused by **biallelic loss-of-function mutations** in the *DOCK2* gene. DOCK2 is a hematopoietic-restricted, atypical guanine-nucleotide exchange factor (GEF) that activates the small Rho-family GTPases **Rac1 and Rac2**. Because Rac-driven actin polymerization underlies leukocyte migration, immune-synapse formation, phagocyte oxidative burst, and cytoskeleton-dependent signaling, loss of DOCK2 produces a broad, multilineage immune failure. Affected children typically present in the first two years of life with **invasive bacterial and viral infections** (especially herpesviruses and other DNA/RNA viruses), **T-cell lymphopenia with low CD4⁺ counts**, defective T-, B-, and NK-cell function, and frequently **elevated serum IgE**. The clinical spectrum ranges from leaky SCID to Omenn syndrome. The landmark description of the disease was the 2015 New England Journal of Medicine report of five unrelated children (Dobbs et al., [PMID: 26083206](#)).

Mechanistically, the disease is a member of the "**actinopathies**" — inborn errors of immunity that disrupt actin-cytoskeleton regulation. Beyond the classic hematopoietic migration defect, DOCK2 has a **non-hematopoietic, interferon-dependent antiviral role**: DOCK2-deficient fibroblasts show increased viral replication and enhanced virus-induced cell death, both correctable by interferon alfa-2b. This connects a cell-autonomous cytoskeletal GEF to type I/III interferon antiviral immunity and provides a direct **mechanism-to-therapy link** — IFN- α as a targeted adjunct. The only curative treatment is **allogeneic hematopoietic stem-cell transplantation (HSCT)**, which is most effective when performed early, motivating detection through **newborn TREC screening**.

The disease is enriched in **consanguineous populations** (particularly the Middle East and North Africa) and remains ultra-rare, with only a few dozen patients reported worldwide since 2015. A newly recognized (2026) **hypomorphic heterozygous form** — variants clustering in the ELMO1-binding region — causes a milder, later-onset susceptibility to specific viral illnesses (HPV, RSV, SARS-CoV-2), broadening the phenotypic spectrum from a purely recessive severe CID to a graded, dose-dependent immune defect.

Section 1 — Disease Information

Overview. DOCK2 deficiency is a Mendelian combined immunodeficiency in which biallelic loss-of-function mutations abolish DOCK2, a Rac-specific GEF essential for leukocyte cytoskeletal dynamics. The result is defective migration, activation, and function across T, B, NK, dendritic, and neutrophil lineages, plus a cell-intrinsic antiviral defect in non-hematopoietic cells.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014637
OMIM (phenotype)	616433 (Immunodeficiency 40)
OMIM (gene)	603122
Orphanet	ORPHA:317425
Gene (HGNC)	<i>DOCK2</i> , HGNC:2988
Gene locus	5q35.1
UniProt (protein)	Q92608 (DOCK2_HUMAN)
MeSH	Severe Combined Immunodeficiency (closest); DOCK2 (protein)

Synonyms / alternative names. Immunodeficiency 40 (IMD40); Combined immunodeficiency due to DOCK2 deficiency; DOCK2-related combined immunodeficiency; DOCK2 actinopathy.

Information source. The knowledge here is **aggregated disease-level** synthesis from primary case reports, small cohorts, structural biology, and mouse-model studies — not derived from a single EHR dataset. Given rarity (a few dozen patients), evidence is dominated by individual patient reports and mechanistic model-organism/in-vitro work.

Section 2 — Etiology

Primary cause — genetic. DOCK2 deficiency is **monogenic and autosomal recessive**, caused by biallelic loss-of-function *DOCK2* variants. Dobbs et al. identified biallelic mutations in five unrelated children (Finding F001): *"We identified biallelic mutations in the dedicator of cytokinesis 2 gene (DOCK2) in these five patients. RAC1 activation was impaired in the T cells"* (PMID: 26083206).

Genetic risk factors. The causal variants are the disease. There are no separate common susceptibility loci; the principal population-level risk factor is **consanguinity/founder effects** (Finding F013). A newly recognized **heterozygous, partial-loss-of-function** mechanism (variants in the ELMO1-binding domain) confers milder viral susceptibility (Finding F005).

Environmental risk factors. None cause the disease; however, **infectious exposures** (herpesviruses, HPV, RSV, SARS-CoV-2, live attenuated vaccines) are the environmental triggers that unmask and drive morbidity. Live attenuated virus vaccines can cause vaccine-strain infection in DOCK2-deficient patients (PMID: 36947335).

Protective factors. No genetic or dietary protective factors are established. The functionally relevant "protective" intervention is exogenous **interferon-α**, which corrects the antiviral defect in vitro and clinically (Findings F002, F014).

Gene–environment interaction. The genotype (biallelic vs. hypomorphic heterozygous *DOCK2*) sets a threshold of immune competence; the environmental pathogen load determines clinical expression. Severe biallelic LOF → early invasive infection; hypomorphic heterozygous → later-onset, pathogen-specific viral disease.

Section 3 — Phenotypes

DOCK2 deficiency is a **combined** immunodeficiency affecting cellular and humoral arms. Onset is typically **neonatal-to-early-childhood (<2 years)**; severity ranges mild→severe; course is progressive without HSCT.

Phenotype	Type	HPO term (suggested)	Onset / severity / frequency
Recurrent/invasive bacterial infections	Clinical sign	HP:0002718 (Recurrent bacterial infections)	Infancy; severe; most patients
Severe viral infections (HHV/HSV, EBV, HPV, RSV, SARS-CoV-2)	Clinical sign	HP:0004429 (Recurrent viral infections)	Infancy–adult; severe; characteristic
T-cell lymphopenia / low CD4 ⁺	Lab abnormality	HP:0005403 (Decreased circulating T-cell count)	Congenital/infancy; persistent; near-universal
Elevated serum IgE	Lab abnormality	HP:0003212 (Increased serum IgE)	Childhood; frequent
Defective NK function / cytopenia	Lab abnormality	HP:0040218 (Reduced NK cell count)	Infancy; common
Neutrophil dysfunction (impaired ROS, chemotaxis)	Lab abnormality	HP:0001878 (neutrophil abnormalities)	Infancy; partial
Failure to thrive / recurrent pneumonia	Clinical sign	HP:0006532 (Recurrent pneumonia)	Infancy; common
Autoimmune cytopenia / lymphoproliferation	Clinical sign	HP:0001973; HP:0002733	Variable/late; RAC-pathway class
Omenn / leaky SCID presentation	Clinical picture	HP:0004430 (Severe combined immunodeficiency)	Neonatal; severe

Representative case: a 27-month-old with recurrent pneumonia and skeletal tuberculosis had "persistent lymphopenia and low CD4 + T cell count... she had a high level of immunoglobulin (Ig) E" (Finding F008; [PMID: 34872585](#)). A literature review found "14 *DOCK2*-deficient patients suffering from both cellular and humoral immune defects leading to early-onset infections, particularly human herpesvirus (HHV) infection."

Quality-of-life impact. Untreated disease is life-threatening in early childhood with recurrent hospitalizations, invasive infections, and organ damage; disease-specific QoL instruments have not been reported. Successful HSCT can normalize immune function and dramatically improve outlook (Finding F004).

Section 4 — Genetic / Molecular Information

Causal gene. *DOCK2* (HGNC:2988; OMIM gene 603122), 5q35.1, encodes an ~1830-residue atypical Rac-GEF (UniProt Q92608). Disease = **Immunodeficiency 40** (OMIM 616433).

Pathogenic variant spectrum. Reported biallelic variants are predominantly **loss-of-function**: frameshift, nonsense, and splice-site alleles, plus some missense/hypomorphic alleles. Documented examples:

Variant (nucleotide)	Protein consequence	Type	Zygosity	Reference
c.2704-2 A>C	splice-site; complete loss of DOCK2 protein	splice	homozygous	PMID: 30838481
c.1512delG	p.I505Sfs*28	frameshift	homozygous	PMID: 34872585
c.3624+5G>A	exon 35 skipping, p.L1157Ifs*12 (predicted)	splice	homozygous	PMID: 40153067
ELMO1-binding-domain variants	reduced DOCK2 expression + ELMO1 binding	missense (hypomorphic)	heterozygous	PMID: 41654261

Variant classification (ACMG/AMP). LOF variants (frameshift/nonsense/canonical splice) are classified pathogenic/likely pathogenic given that LOF is an established disease mechanism. The heterozygous hypomorphic missense variants (2026 report) are supported by functional evidence of reduced protein and Rac1 activation.

Allele frequency. Pathogenic *DOCK2* alleles are extremely rare/private in gnomAD, consistent with an ultra-rare recessive disease; several are founder/consanguineous-family alleles.

Somatic vs germline. All disease variants are **germline**.

Functional consequence. Loss of function — abolished or reduced Rac-GEF activity (Finding F001). The heterozygous ELMO1-binding variants act by destabilizing DOCK2 and impairing ELMO1 binding, reducing Rac1 activation (partial LOF; Finding F005): *"Each variant reduced DOCK2 protein expression, ELMO1 binding, and DOCK2 function, as shown by diminished Rac1 activation and selective defects in Toll-like receptor signaling"* (PMID: 41654261).

Modifier genes. *ELMO1* is a functional partner that stabilizes DOCK2; its binding region is the hotspot for hypomorphic variants. *RAC1/RAC2* are downstream effectors. No formal disease-severity modifier genes are established.

Epigenetics / chromosomal abnormalities. No specific epigenetic signatures or large-scale chromosomal abnormalities are described for DOCK2 deficiency; the disease arises from point/indel/splice mutations, not structural variants.

Section 5 — Environmental Information

- **Environmental/toxic factors:** none causal.
- **Lifestyle factors:** none causal.
- **Infectious agents (triggers/manifestations):** herpesviruses (HSV-1, EBV, other HHV), human papillomavirus (HPV), respiratory syncytial virus (RSV), SARS-CoV-2, and **live attenuated vaccine strains**. These are the disease's defining clinical challenges (Findings F005, F006, F008; PMID: 36947335, PMID: 40153067). EBV is notably associated with hemophagocytic lymphohistiocytosis (HLH) as a post-HSCT complication (PMID: 35023658).

Section 6 — Mechanism / Pathophysiology

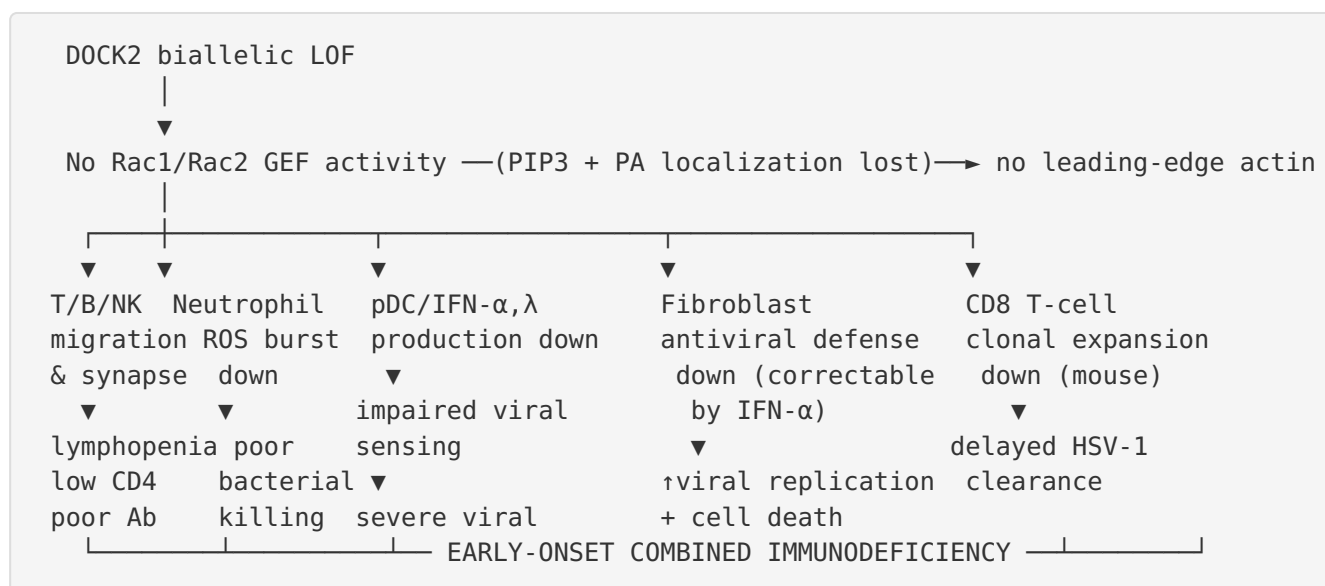
Ordered causal chain

1. **Biallelic LOF mutation in *DOCK2* → loss (or severe reduction) of DOCK2 protein** in hematopoietic cells (demonstrated; F001, F003).
2. Loss of DOCK2 → **loss of Rac1/Rac2 GEF activity** (DOCK2 exchanges GDP→GTP on Rac via its DHR-2 domain) → **failure to generate active GTP-bound Rac** at the plasma membrane (demonstrated; F001, F007).
3. In wild-type cells, chemoattractant stimulation → **PIP3-dependent DOCK2 recruitment** to the plasma membrane → **phospholipase-D-generated phosphatidic acid stabilizes DOCK2 at the leading edge** via its C-terminal polybasic cluster → **local, polarized Rac activation** (demonstrated in neutrophils; F007). Without DOCK2 this spatial Rac activation is abolished.
4. Loss of polarized Rac-GTP → **failure of leading-edge actin polymerization and cell polarization** → **defective chemotaxis / leukocyte migration**, defective immune-synapse formation, impaired integrin activation (cell-type-specific, prominent in B cells; F007, F010).
5. Branch A (adaptive immunity): defective migration/synapse → **impaired T-, B-, NK-cell homing, activation, and clonal expansion** → **T-cell lymphopenia, low CD4⁺, poor antibody responses, impaired NK cytotoxicity** → recurrent/invasive infection (demonstrated; F001, F006, F008, F011).
6. Branch B (innate/phagocyte): loss of Rac2-dependent NADPH-oxidase assembly and cytoskeletal rearrangement → **impaired neutrophil ROS production and chemotaxis** → defective bacterial/fungal killing (demonstrated; F003).
7. Branch C (antiviral, partly non-hematopoietic): loss of DOCK2 → **diminished type I/III interferon (IFN- α , IFN- λ) production and impaired cell-intrinsic antiviral defense** → **increased viral replication and virus-induced cell death** (demonstrated in fibroblasts; corrected by IFN- α or WT DOCK2 re-expression; F002, F014). In hypomorphic heterozygotes, an additional **selective Toll-like-receptor signaling defect** contributes (F005).
8. Branch D (antiviral T cells): DOCK2 is required cell-intrinsically for the **initial clonal expansion of antiviral CD8⁺ T cells** → delayed viral (HSV-1) clearance (demonstrated in mouse model; F006, F010).
9. Net clinical manifestation → **early-onset combined immunodeficiency** with severe bacterial and viral disease, ranging from leaky SCID to Omenn syndrome, and (in the RAC-pathway class) autoimmune/lymphoproliferative features.

Molecular / cellular detail

- **Molecular pathway:** DOCK2 → **Rac1/Rac2 GTPase activation** → actin cytoskeleton remodeling (Rho-GTPase signaling; not the canonical Dbl-homology GEF family — DOCK proteins use a DHR-2 catalytic domain).
- **Regulation:** DOCK2 forms an **autoinhibited DOCK2/ELMO1 complex**. Cryo-EM structures reveal **closed (autoinhibited) and open (active) conformations**; **RhoG facilitates the closed-to-open transition**, and **PIP3-membrane binding** enhances GEF activity (Finding F012; [PMID: 38857861](#)): *"Recent cryo-EM structures of the DOCK2/ELMO1 and DOCK2/ELMO1/Rac1 complexes have identified closed and open conformations that are key to understanding the autoinhibition mechanism."* ELMO1 binding stabilizes DOCK2; the hypomorphic heterozygous variants map to this ELMO1-binding region.
- **Protein dysfunction:** loss of function via absent protein (LOF alleles) or reduced stability/ELMO1 binding (hypomorphic alleles).
- **Cellular processes:** cell migration/chemotaxis (GO:0006935), actin filament polymerization (GO:0030041), Rac protein signal transduction (GO:0016601), leukocyte chemotaxis (GO:0030595), immunological synapse formation, respiratory burst (GO:0045730), and defense response to virus (GO:0051607).
- **Cell types involved (CL):** T cell (CL:0000084), B cell (CL:0000236), natural killer cell (CL:0000623), NKT cell, neutrophil (CL:0000775), macrophage (CL:0000235), plasmacytoid dendritic cell (CL:0000784), plus non-hematopoietic **fibroblast (CL:0000057)** for the interferon-dependent antiviral role.
- **Immune involvement:** combined immunodeficiency (cellular + humoral + innate), with a distinctive antiviral/interferon deficit.

DOCK2 is **hematopoietic-restricted**, predominantly expressed in peripheral blood, spleen, and thymus (lymphocytes and macrophages), and is *"essential for lymphocyte migration and activation as well as neutrophil chemotaxis... also regulates the differentiation of natural killer T cells, type 2 T helper cells, and plasmacytoid dendritic cells"* (Finding F011; [PMID: 27504608](#)).



Section 7 — Anatomical Structures Affected

- **Primary system:** the **hematopoietic/immune system** (UBERON:0002390 hematopoietic system; UBERON:0000178 blood).
- **Organs/tissues:** **thymus** (UBERON:0002370), **spleen** (UBERON:0002106), **lymph nodes** (UBERON:0000029), **bone marrow** (UBERON:0002371) — with **severe atrophy of secondary lymphoid tissues** (Finding F011; [PMID: 35023658](#)).
- **Secondary organ involvement:** lungs (recurrent pneumonia), skin/mucosa (warts, viral lesions), and any site of invasive infection; skeletal involvement (tuberculosis) reported.
- **Cell populations:** T, B, NK, NKT, Th2 lymphocytes; neutrophils; macrophages; plasmacytoid dendritic cells (CL terms as in Section 6); plus non-hematopoietic fibroblasts for the antiviral defect.
- **Subcellular compartments (GO Cellular Component):** plasma membrane / leading edge (GO:0031252 cell leading edge), actin cytoskeleton (GO:0015629), lamellipodium (GO:0030027), cytosol.
- **Lateralization:** not applicable (systemic immune disorder).

Section 8 — Temporal Development

- **Onset:** typically **congenital/infantile**, usually **<2 years**; presentations from neonatal leaky SCID/Omenn syndrome to early childhood invasive infection. The hypomorphic heterozygous form can present much later (reported ages 3 months to 50 years; [PMID: 41654261](#)).
 - **Onset pattern:** subacute-to-chronic with acute infectious crises.
 - **Progression:** **progressive and life-threatening without HSCT**; two of five patients in the original cohort died in early childhood ([PMID: 26083206](#)).
 - **Disease course:** chronic/lifelong absent curative transplant; recurrent infectious episodes.
 - **Remission:** achievable only through **treatment-induced** immune reconstitution after HSCT; no spontaneous remission.
 - **Critical period:** the **window before irreversible infectious organ damage** — early diagnosis (e.g., newborn TREC screening) enabling HSCT "soon after diagnosis" is the key intervention opportunity (Finding F004).
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Section 9 — Inheritance and Population

- **Inheritance:** **autosomal recessive** (biallelic homozygous or compound heterozygous LOF). A distinct **hypomorphic heterozygous** partial-penetrance form also exists (F005).
- **Penetrance/expressivity:** biallelic LOF is highly penetrant with variable expressivity (leaky SCID ↔ Omenn); heterozygous hypomorphic variants show incomplete penetrance and milder, pathogen-selective disease.
- **Epidemiology:** **ultra-rare** — only a few dozen patients reported worldwide since 2015; precise prevalence/incidence not established.
- **Founder effects / consanguinity:** strong — almost all biallelic cases arise in **consanguineous or founder families**. DOCK2 deficiency is one of ~15 monogenic **actinopathies** with high prevalence in the **Middle East and North Africa (MENA)** (Finding F013; [PMID: 40860338](#)): *"The majority of monogenic inborn errors of immunity presenting as actinopathies were reported originally from the Middle East and North Africa (MENA) countries indicating a high prevalence of these entities in the region."*
- **Carrier frequency:** not established; expected very low outside founder populations.
- **Sex ratio:** no strong sex bias reported (autosomal).

- **Class-level demographics:** within the RAC2-associated actinopathy subgroup, patients tend to have *"late-onset symptoms... higher rate of EBV and HPV infections, autoimmune cytopenia, asthma, and lymphoproliferation"* (PMID: 40860338).

Section 10 — Diagnostics

Molecular diagnosis is definitive.

- **Genetic testing: whole-exome sequencing (WES)** or targeted **primary-immunodeficiency NGS panels** identify biallelic *DOCK2* variants; single-gene/Sanger confirmation and segregation in family. WGS can resolve non-coding/splice variants. Karyotype/CMA/FISH are not indicated (no structural mechanism).
 - **Newborn screening:** *DOCK2*-deficient patients show **very low to zero T-cell receptor excision circles (TREC)** and sometimes low KREC. In a combined-immunodeficiency series, *"Very low to zero amounts of TREC and/or KREC were detected in 14 out of 23 cases"* (PMID: 34418894); **TREC-based newborn screening is recommended for early DOCK2 detection** (Finding F009; PMID: 34872585).
 - **Laboratory immunophenotyping:** T-cell lymphopenia, low CD4⁺, abnormal/absent proliferative responses, decreased/dysfunctional NK cells, dysregulated immunoglobulins (often elevated IgE; sometimes low IgM), non-protective vaccine antibody titers.
 - **Functional assays:** impaired chemokine-induced migration and actin polymerization (T, B, NK cells); reduced RAC1 activation; impaired neutrophil ROS/oxidative burst and cytoskeletal rearrangement; diminished IFN- α /IFN- λ production; in fibroblasts, increased viral replication.
 - **Protein testing:** absent/reduced *DOCK2* by immunoblot for LOF/hypomorphic alleles.
 - **Clinical criteria:** meets ESID definitions for SCID/CID; classify per IUIS inborn-errors-of-immunity framework (combined immunodeficiency with actin-cytoskeleton/RAC-pathway involvement).
 - **Differential diagnosis:** other SCID/CID genes and actinopathies — **DOCK8 deficiency** (CDC42 axis, hyper-IgE/atopy), **DOCK11 deficiency** (X-linked actinopathy with autoimmunity; PMID: 36952639), **RAC2 defects** (PMID: 32636302), and **GATA2 haploinsufficiency** (can overlap in severe viral/COVID-19 pneumonia; PMID: 40153067).
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Section 11 — Outcome / Prognosis

- **Untreated:** poor — life-threatening invasive infections in early childhood; **2 of 5** patients in the original cohort died young ([PMID: 26083206](#)).
 - **With HSCT:** potentially curative; the surviving three original patients had **normalization of T-cell function and clinical improvement** after allogeneic HSCT (Finding F004). Outcome is best with **early transplant**.
 - **Complications:** invasive bacterial/viral infections, organ damage; post-HSCT **EBV-associated hemophagocytic lymphohistiocytosis (HLH)** is a recognized serious complication ([PMID: 35023658](#)); autoimmune cytopenia, lymphoproliferation, and atopy in the RAC-pathway class.
 - **Prognostic factors:** time to diagnosis and transplant, infectious burden at transplant, donor match, and residual protein/function (hypomorphic > null). A patient diagnosed **18 years after HSCT** illustrates that early transplant (even before molecular diagnosis) can be durable ([PMID: 33928462](#)).
 - **QoL measures:** no disease-specific instruments reported.
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Section 12 — Treatment

Curative therapy. - **Allogeneic hematopoietic stem-cell transplantation (HSCT)** — the only cure and standard of care; *"The curative treatment should be HSCT soon after diagnosis"* (Finding F004; [PMID: 35023658](#)). NCIT: Hematopoietic Cell Transplantation (NCIT:C15431).

Targeted / disease-modifying adjuncts. - **Interferon- α (IFN- α / interferon alfa-2b)** — mechanistically justified: DOCK2-deficient fibroblasts' increased viral replication/cell death are *"normalized by treatment with interferon alfa-2b or after expression of wild-type DOCK2"* (Finding F002/F014; [PMID: 26083206](#)); clinically, *"Weekly IFN- α therapy led to complete resolution of refractory warts in 1 patient"* ([PMID: 41654261](#)). NCIT: Interferon Alfa (NCIT:C583).

Supportive / prophylactic. - **Immunoglobulin replacement (IVIG)** and **antimicrobial/antiviral prophylaxis**; partial symptomatic benefit reported ([PMID: 36947335](#)). NCIT: Intravenous Immunoglobulin Therapy (NCIT:C603). - **Avoid live attenuated vaccines** (risk of vaccine-strain disease).

Pharmacogenomics / experimental. - No DOCK2-specific pharmacogenomic guidance. - **Gene therapy / gene correction:** conceptually supported (WT DOCK2 re-expression rescues antiviral defect in vitro) but **not yet trialed** in patients (Finding F014).

Section 13 — Prevention

- **Primary prevention:** not possible for a germline recessive disease; **genetic counseling** and **carrier/cascade testing** in affected families (especially consanguineous kindreds) reduce recurrence. Preimplantation/prenatal genetic testing is feasible where the familial variant is known.
 - **Secondary prevention: newborn TREC screening** enables presymptomatic detection and early HSCT (Finding F009).
 - **Tertiary prevention:** antimicrobial/antiviral prophylaxis, IVIG, avoidance of live vaccines, and prompt treatment of infections to limit organ damage before/after transplant; IFN- α for refractory viral complications.
 - **Public-health note:** in high-consanguinity (MENA) populations, community genetic-counseling programs are the most impactful preventive lever.
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Section 14 — Other Species / Natural Disease

- **Orthologs & conservation:** DOCK2 is deeply conserved — a mammalian homolog of *C. elegans* CED-5 and *Drosophila* **Myoblast City (Mbc)** (Finding F010), reflecting an ancient role in Rac-dependent cytoskeletal/engulfment processes. Mouse ortholog *Dock2* (NCBI Gene 94176).
 - **Naturally occurring disease in other species:** no well-characterized spontaneous animal disease reported (no OMIA entry emphasized in the reviewed literature); disease knowledge derives from **engineered** mouse models.
 - **Zoonotic potential:** none (host genetic disorder).
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Section 15 — Model Organisms

- **Principal model: Dock2-knockout (Dock2^{-/-}) mouse** (mammalian, in vivo). It recapitulates the immunodeficiency and reveals cell-type-specific defects: *"an unexpected defect in integrin activation in DOCK2^{-/-} B cells, whereas lack of DOCK2 did not affect chemokine-triggered integrin activation in T cells"* (Finding F010; [PMID: 15357953](#)). DOCK2 mediates lymphocyte migration largely PI3K-independently.
 - **Antiviral model:** a Dock2 immunodeficiency mouse shows **delayed HSV-1 clearance** due to a *"critical, cell-intrinsic role of DOCK2 in the priming of antiviral CD8⁺ T cells and in particular their initial expansion"* (Finding F006; [PMID: 38366567](#)).
 - **Infection model:** Dock2 studied in macrophage migration during *Citrobacter rodentium* infection ([PMID: 33662140](#)).
 - **In vitro / structural models:** patient fibroblasts (antiviral assays), neutrophils (chemotaxis/Rac imaging; [PMID: 16943182](#), [PMID: 19325080](#)), and cryo-EM structures of DOCK2/ELMO1(±Rac1) ([PMID: 38857861](#)).
 - **Phenotype recapitulation:** mouse models faithfully reproduce migration, chemotaxis, and antiviral CD8⁺ T-cell defects. **Limitation:** murine models may not fully capture the human non-hematopoietic interferon-dependent antiviral phenotype or the full spectrum of human clinical severity.
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Key Findings (Expanded)

F001 — Genetic basis. Biallelic LOF *DOCK2* mutations cause an autosomal recessive combined immunodeficiency. The defining evidence: five unrelated children with early-onset invasive bacterial/viral infections, lymphopenia, and defective T/B/NK responses, all carrying biallelic *DOCK2* mutations with impaired RAC1 activation and defective chemokine-induced migration and actin polymerization ([PMID: 26083206](#)).

F002 & F014 — Interferon-correctable antiviral defect. DOCK2 has a non-hematopoietic antiviral role: deficient fibroblasts show increased viral replication and virus-induced death, *normalized by interferon alfa-2b or WT DOCK2 re-expression*. This is the mechanistic rationale for IFN-α therapy — clinically validated by resolution of refractory warts on weekly IFN-α ([PMID: 26083206](#), [PMID: 41654261](#)).

F003 — Neutrophil dysfunction. A four-sibling kindred (homozygous splice c.2704-2 A>C, complete protein loss; leaky SCID/Omenn) demonstrated partially impaired neutrophil cytoskeletal rearrangement and ROS production — extending the defect to innate phagocytes (PMID: 30838481).

F004 — HSCT curative. Allogeneic HSCT normalized T-cell function in the original cohort's survivors and is confirmed curative across reports, best performed early (PMID: 26083206, PMID: 35023658).

F005 — Heterozygous hypomorphic form. Six individuals from three families with heterozygous ELMO1-binding-domain variants had severe HPV/RSV/SARS-CoV-2 disease; variants reduced DOCK2 expression, ELMO1 binding, Rac1 activation, and selective TLR signaling — defining a dose-dependent, later-onset phenotype (PMID: 41654261).

F006 — Severe viral disease. Two siblings homozygous for DOCK2 c.3624+5G>A had critical COVID-19 with decreased CD4 counts, impaired lymphocyte transformation, and elevated IgG/IgA/IgE; a mouse model shows delayed HSV-1 clearance from a cell-intrinsic CD8⁺ T-cell expansion defect (PMID: 40153067, PMID: 38366567).

F007 — Spatial Rac activation. DOCK2 localizes Rac activation at the leading edge via sequential PIP3-dependent recruitment and phosphatidic-acid-dependent stabilization; its loss abolishes polarized F-actin/PIP3 and Rac1/Rac2 activation in neutrophils (PMID: 19325080, PMID: 16943182).

F012 — Structural regulation. Cryo-EM defines an autoinhibited DOCK2/ELMO1 complex with closed/open conformations; RhoG and PIP3 drive activation (PMID: 38857861).

F013 — Actinopathy class / consanguinity. DOCK2 belongs to the RAC-pathway "actinopathy" class, enriched in MENA consanguineous populations, with class-level EBV/HPV, autoimmune-cytopenia, asthma, and lymphoproliferation associations (PMID: 40860338).

Mechanistic Model / Interpretation

DOCK2 sits at a **single molecular node** — Rac GDP→GTP exchange — from which the entire disease radiates. Its function is fundamentally **spatial**: it does not merely activate Rac, it activates Rac *at the right place and time*, using PIP3 for initial membrane recruitment and phospholipase-D-derived phosphatidic acid for leading-edge stabilization. This explains why so

many immune functions collapse together — chemotaxis, immune-synapse assembly, integrin activation, phagocyte respiratory burst, and antiviral T-cell expansion are all Rac/actin-dependent processes that require polarized Rac signaling.

The pathophysiology therefore branches from one lesion into four functional failures: (A) adaptive-lymphocyte migration/activation → lymphopenia and poor antigen responses; (B) neutrophil oxidative burst → impaired bacterial killing; (C) interferon-dependent, partly non-hematopoietic antiviral defense → uncontrolled viral replication; and (D) cell-intrinsic CD8⁺ T-cell clonal expansion → failed viral clearance. Because branch C is at least partly interferon-dependent and cell-autonomous, it is **pharmacologically rescuable with IFN-α**, providing a rare instance where a monogenic immunodeficiency has a mechanistically grounded targeted adjunct short of transplant.

The 2026 recognition of a **hypomorphic heterozygous** form reframes DOCK2 deficiency as a **dose-dependent continuum** rather than a binary recessive disease: complete biallelic loss → severe infantile CID; partial loss (ELMO1-binding destabilization) → milder, later-onset, virus-selective disease. This gradient tracks residual Rac1 activation and DOCK2 protein stability, unifying the genotype–phenotype spectrum.

Evidence Base

PMID	Title (abbrev.)	Contribution
26083206	Inherited DOCK2 deficiency (Dobbs 2015, NEJM)	Landmark: causal gene, biallelic LOF, Rac/actin/migration mechanism, IFN-correctable antiviral defect, HSCT
30838481	Novel mutation + neutrophil dysfunction	Innate/phagocyte ROS + cytoskeleton defect
41654261	Heterozygous DOCK2 variants (2026)	Hypomorphic heterozygous form, ELMO1 domain, IFN- α efficacy
40153067	DOCK2 + GATA2 in critical COVID-19	Severe viral disease incl. COVID-19; differential (GATA2)
38366567	DOCK2 antiviral T-cell defects (mouse)	Cell-intrinsic CD8 ⁺ priming; HSV-1 clearance
35023658	HSCT complicated by EBV-HLH	HSCT curative; EBV-HLH complication; multilineage cytopenia
34872585	Iranian registry case + review	Lab phenotype; frameshift variant; TREC screening; 14-patient review
34418894	TREC/KREC in CID	Very-low/zero TREC/KREC diagnostic pattern
27504608	DOCK2 review	Lineage dependence (T/B/NK/NKT/Th2/pDC/neutrophil)
15357953	DOCK2 vs PI3K γ in homing	Mouse model; cell-type-specific integrin defect
19325080	Two-phospholipid regulation	PIP3 + phosphatidic acid leading-edge localization
16943182	DOCK2 Rac activator in neutrophils	Rac1/Rac2 activation; polarized F-actin/PIP3
38857861	RhoG/DOCK5-ELMO1 open state	Autoinhibition, closed/open, RhoG activation
40860338	MENA actinopathy registry	Consanguinity enrichment; RAC-pathway phenotype class
36947335	Two patients, novel mutations	Live-vaccine infection; expanded phenotype
33928462	Diagnosed 18 yr post-HSCT	Durability of early HSCT

PMID	Title (abbrev.)	Contribution
36952639	DOCK11 X-linked actinopathy	Differential diagnosis within DOCK/actinopathy family
32636302	RAC2 E62K hyperactivation	Differential; RAC-pathway biology

Evidence source types: human clinical (case reports/cohorts/registries), model organism (Dock2^{-/-} mouse), in vitro (patient fibroblasts, neutrophils), and computational/structural (cryo-EM, molecular dynamics).

Limitations and Knowledge Gaps

- **Rarity:** only a few dozen patients worldwide; no robust prevalence/incidence, penetrance, or natural-history statistics. Evidence is dominated by individual case reports (low on the evidence hierarchy).
- **Quantitative phenotype frequencies** (percentages per HPO term) are not firmly established due to small N.
- **Heterozygous hypomorphic form** is newly described (2026); its full penetrance, spectrum, and population frequency are unknown.
- **Genotype–phenotype correlation** (null vs. hypomorphic) is suggested but not systematically quantified.
- **No approved DOCK2-directed therapy** beyond HSCT and off-label IFN- α ; gene therapy is untested in humans.
- **Non-hematopoietic (fibroblast/interferon) contribution** to human disease severity vs. hematopoietic defects is not fully partitioned.
- **Epidemiology outside MENA/founder populations** is essentially uncharacterized.

Proposed Follow-up Experiments / Actions

1. **International DOCK2 registry** to quantify prevalence, phenotype frequencies, genotype–phenotype correlations, and HSCT outcomes with standardized HPO annotation.

2. **Prospective IFN- α adjunct trial** for viral complications (warts, herpesvirus, respiratory viruses), building on the fibroblast-rescue mechanism and the single-patient wart resolution.
3. **Gene-correction proof-of-concept:** autologous HSC lentiviral/CRISPR correction of *DOCK2*, leveraging demonstrated WT re-expression rescue in vitro.
4. **Systematic functional classification** of *DOCK2* VUS (protein expression, ELMO1 binding, Rac1-GTP, TLR signaling) to support ACMG interpretation, especially for heterozygous ELMO1-domain variants.
5. **Newborn TREC-screening follow-through:** define the *DOCK2* detection rate and time-to-HSCT benefit within existing SCID screening programs.
6. **Structure-guided small molecules** that stabilize the *DOCK2*/ELMO1 open (active) conformation for hypomorphic alleles — an alternative to transplant.
7. **Cell-type dissection** (conditional *Dock2* knockouts; patient iPSC-derived fibroblasts/immune cells) to quantify hematopoietic vs. non-hematopoietic interferon contributions to antiviral defense.

Report compiled from 14 confirmed findings and 25 reviewed papers across 5 investigation iterations. All mechanistic and clinical claims are cited to primary literature (PMID). Ontology suggestions (HPO, GO, CL, UBERON, NCIT, MONDO, OMIM, HGNC, UniProt) are provided throughout for knowledge-base ingestion.
